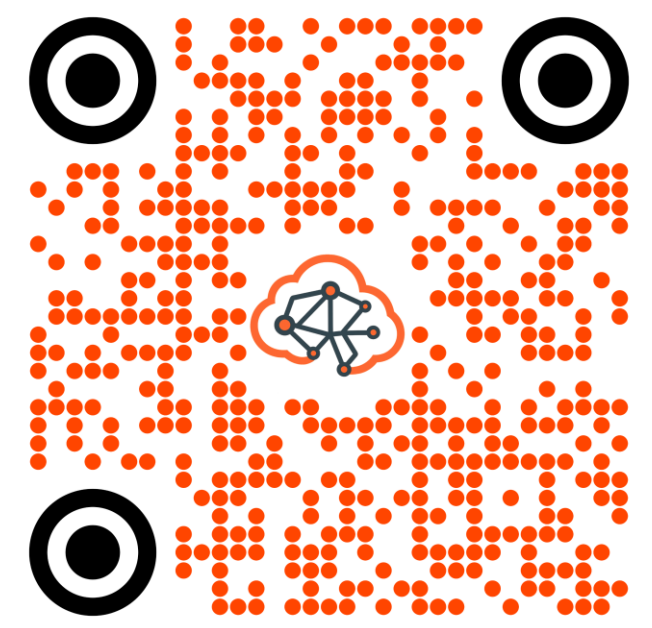
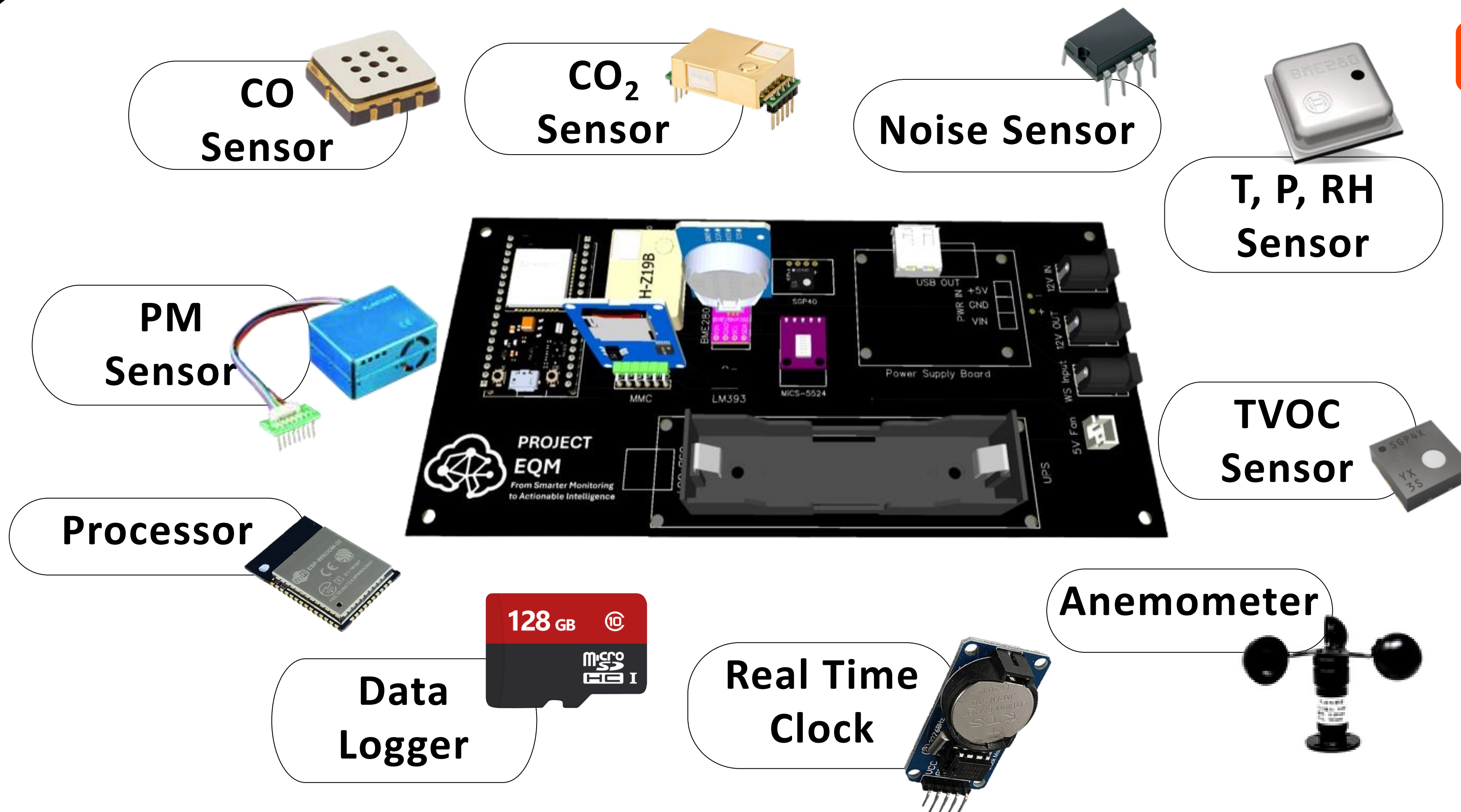


- Focuses on the design and deployment of an **integrated low-cost sensor network** for comprehensive monitoring of PM, CO, CO<sub>2</sub>, noise, temperature, humidity, wind, and TVOCs.
- Facilitates **city-wide, real-time environmental data acquisition** through a centralized server, enabling seamless monitoring across urban landscapes.
- Utilizes a **hybrid CNN-BiLSTM framework** for anomaly detection, spatio-temporal analysis, and multi-scale forecasting (24-hour, 7-day, 1-month) of air quality dynamics.
- Translates complex environmental data into **actionable insights**, embodying the motto: **FROM SMARTER MONITORING TO ACTIONABLE INTELLIGENCE**.



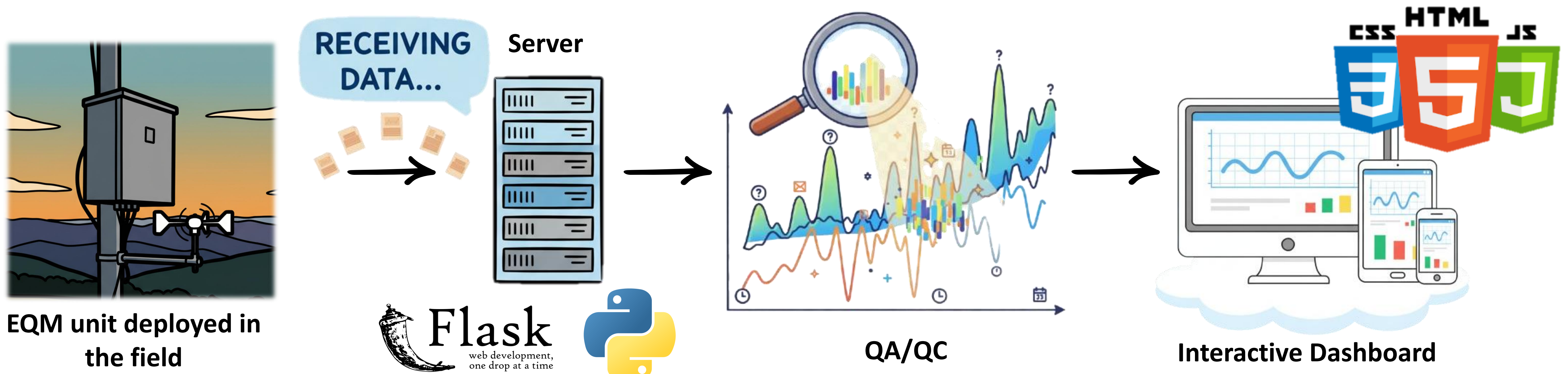
Scan for  
Dashboard



### DIFFERENT COMPONENTS OF EQM

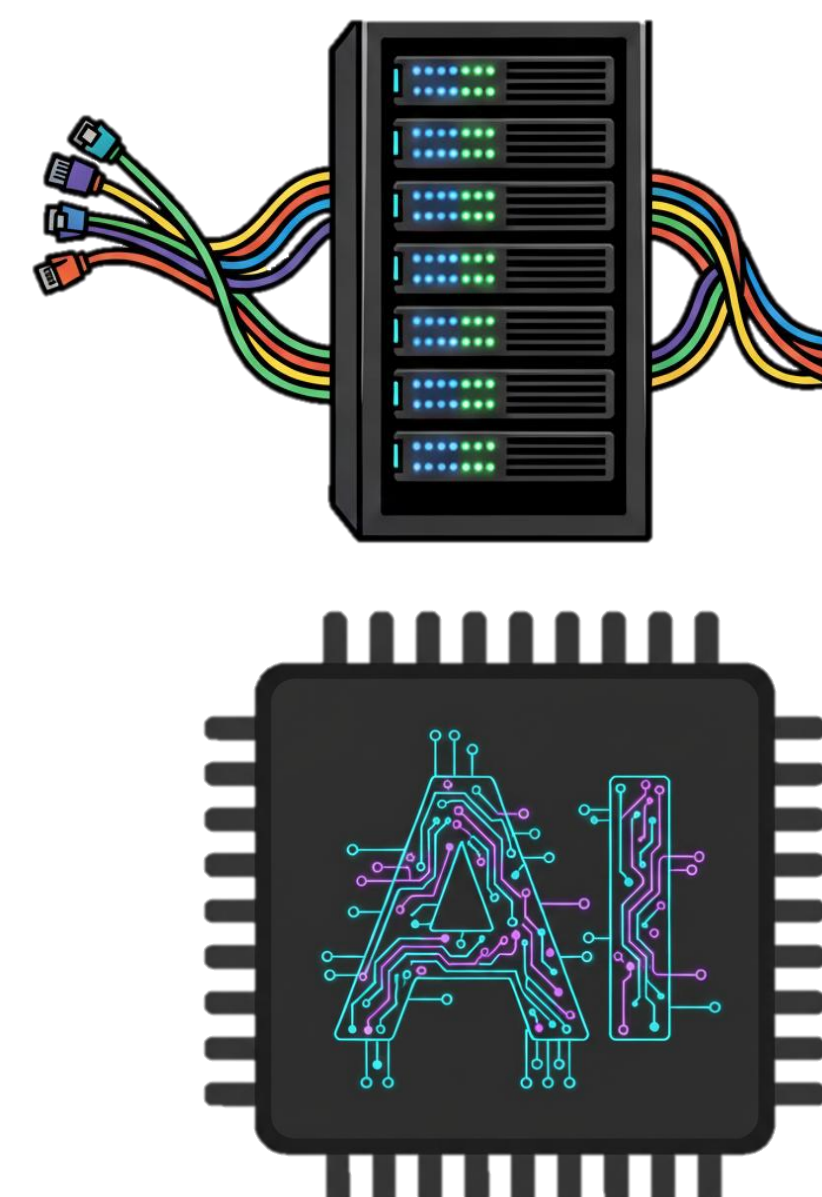
- PM Sensor** – Measures PM<sub>1</sub>, PM<sub>2.5</sub>, PM<sub>10</sub> concentrations.
- CO<sub>2</sub> Sensor** – Monitors carbon dioxide for air quality and ventilation.
- CO Sensor** – Detects carbon monoxide levels.
- TVOC Sensor** – Detects total volatile organic compounds.
- T, P, RH Sensor** – Measures temperature, pressure, and relative humidity.
- Anemometer** – Measures wind speed and direction.
- Noise Sensor** – Records ambient noise for pollution analysis.
- Processor** – Integrates sensors, processes data, and manages communication.
- Real Time Clock** – Provides accurate timestamps.
- Data Logger** – Stores large sensor datasets.

### NETWORK ARCHITECTURE



### Server

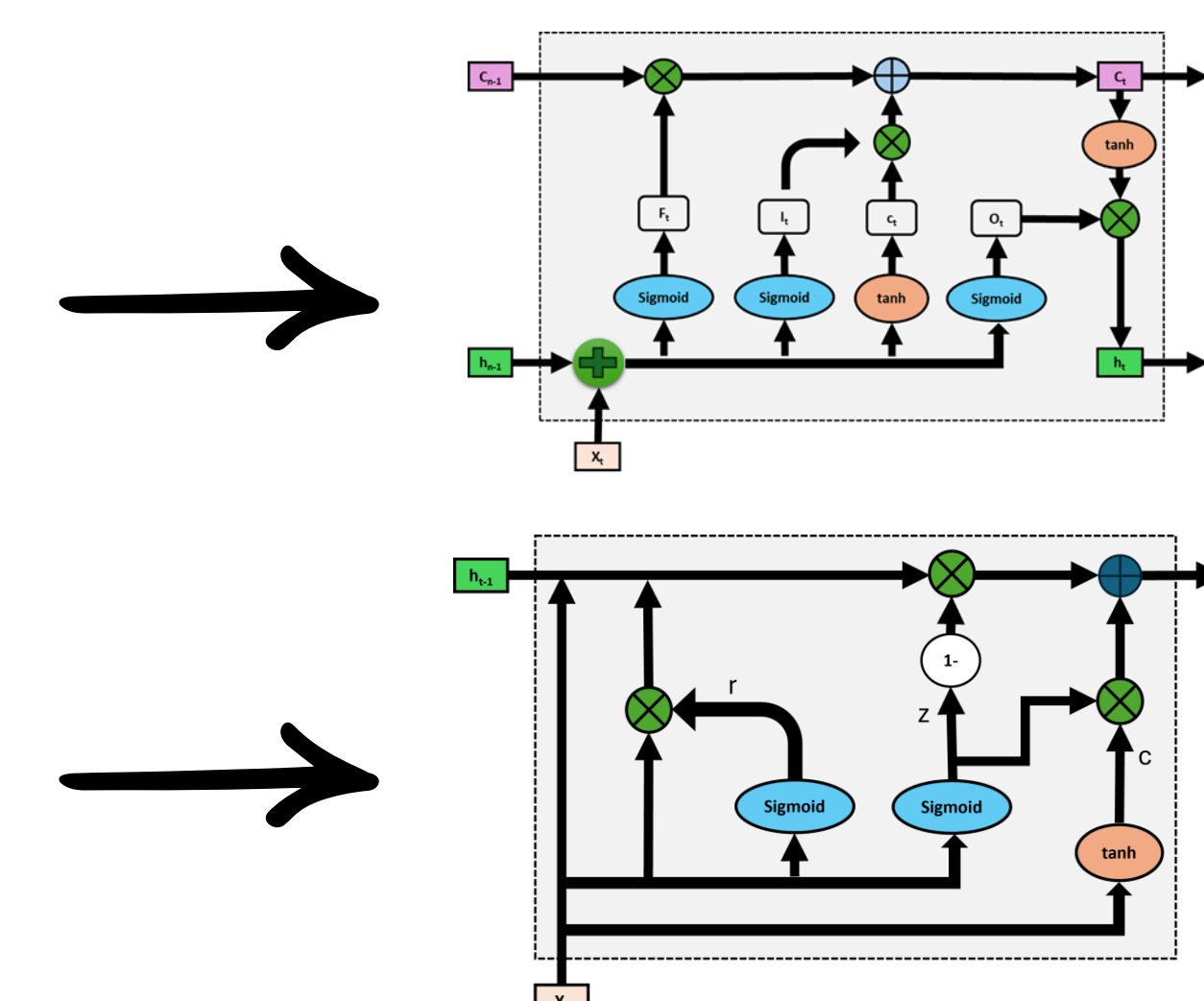
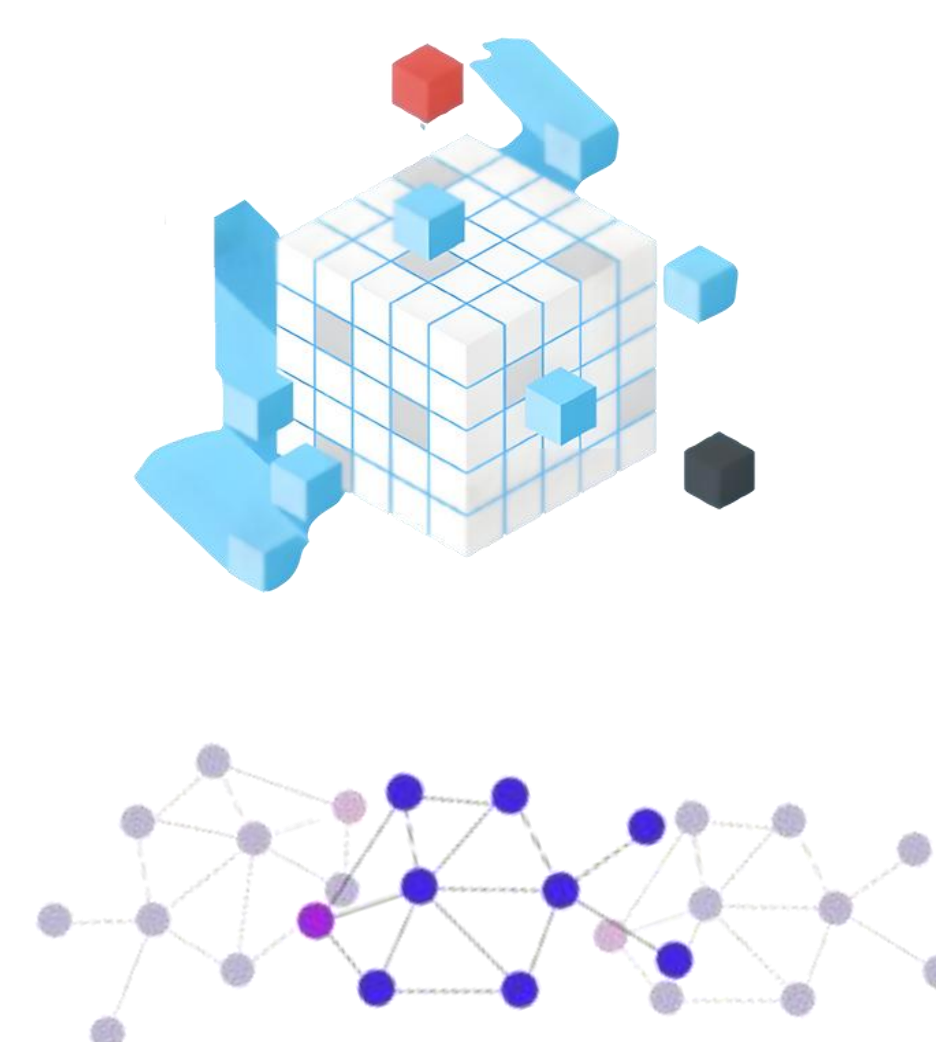
- CNN-BiLSTM / GNN-GRU:** Capture spatial + temporal patterns for prediction.
- Spatiotemporal Processing:** Combine location & time data for analysis.
- Anomaly Detection:** Spot sudden spikes, drops, or sensor errors.



CNN

GNN

### DATA ANALYSIS



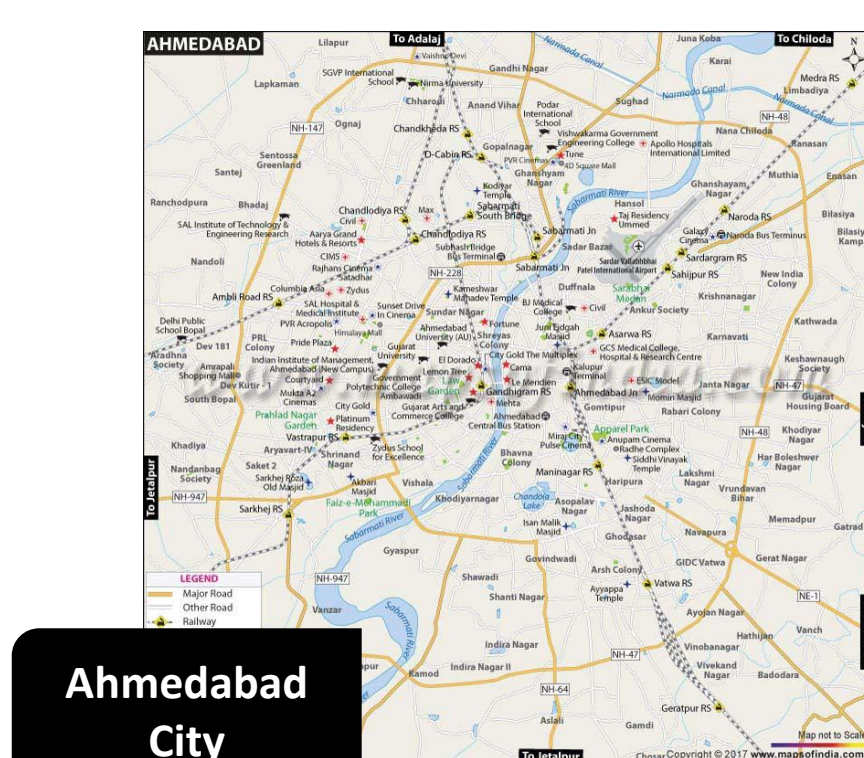
LSTM

GRU

OUTPUT

### DEPLOYMENTS

- Initial trial deployment planned at IITGN campus for testing and validation.
- Large-scale deployment of 50–60 EQM units across Ahmedabad for city-wide, real-time monitoring.
- Strengthens environmental data collection by providing continuous and distributed measurements.
- Enables monitoring of air quality, noise levels, and other key environmental parameters for better urban management.



Project Collaborator:  
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